INFO 5368-030:Final Project Midpoint Check-in

Modeling Pit Stop Strategy in Formula 1: A Machine Learning Approach

He Sun, Yilin Wang, Weifan Wu, Peter Ye

1. Project Overview

This project develops a machine learning system to predict whether a Formula 1 driver will pit on the next lap using lap-level race data. Our current project includes two offline models, a Streamlit-based front end, and supporting visualizations for data exploration and prediction analysis.

2. Progress

URL:

`https://github.com/ww566-wwf/INFO-5368-PAML-Project-Team-ORDS.git`

Sketches of Streamlit UI:

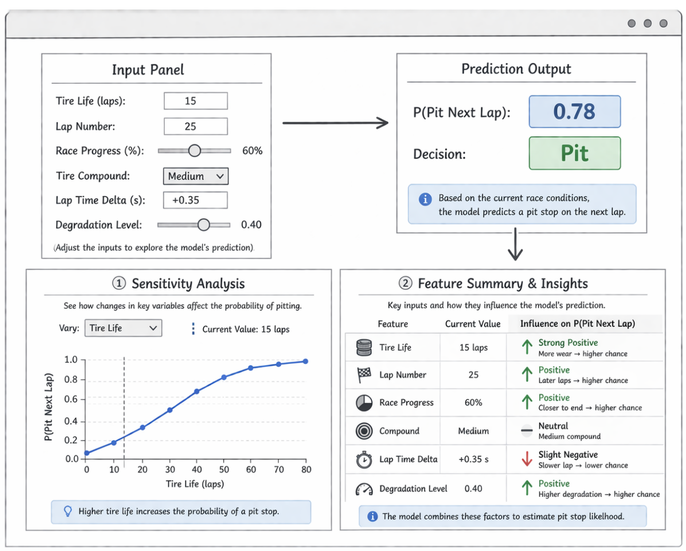

Figure 1: Current Streamlit interface design for pit-stop prediction.

Instructions to Run Streamlit Code:

1. Clone the project repository.
2. Install required packages from `requirements.txt`.
3. Run the Streamlit application in terminal with: streamlit run app.py
4. Open the local URL generated by Streamlit in a browser.

Offline / Online Experiments Completed:

- We completed the main preprocessing pipeline for the lap-level Formula 1 dataset, including data inspection, missing-value handling, outlier treatment, feature engineering, categorical encoding, and train-set-based standardization.
- We identified an anomalous pit-stop labeling pattern in the 2023 subset and excluded that season from the modeling dataset. We then finalized a time-based split with 2022 and 2024 for training and 2025 as the held-out test set.
- We implemented and trained two offline models in NumPy from scratch: logistic regression as a baseline and a one-hidden-layer neural network. We evaluated both models using precision, recall, F1-score, and ROC-AUC.
- We completed a threshold-tuning experiment using the 2024 validation fold and compared default versus tuned thresholds on the 2025 test set. The current best-performing model is the one-hidden-layer ANN with a tuned threshold.

Website Development Completed:

- We completed the current Streamlit interface sketch, as shown in Figure 1. The layout includes the main input, prediction, and analysis panels for interactive pit-stop prediction.
- We are currently integrating the model-training pipeline with the Streamlit application logic. In the final version, the app will load the saved model weights, feature order, scaling parameters, driver mapping, and configuration files so that user inputs are transformed into the same feature format used during training before generating real-time predictions.

Scope and Goal Revisions:

The main goal of the project has remained the same: predicting whether a Formula 1 driver will pit on the next lap using lap-level race data. However, we revised the scope in several practical ways based on the work completed so far. We excluded the 2023 season after identifying an anomalous pit-stop labeling pattern, narrowed the modeling focus to logistic regression and a one-hidden-layer neural network implemented in NumPy, and refined the Streamlit design toward a more interactive prediction-oriented interface. These changes were made to improve data quality, keep the project technically manageable, and better support the final deployed application.

3. Remaining Work and Plan

The main work remaining is to finalize model selection and complete full Streamlit integration. We still need to finish hyperparameter tuning, confirm the final deployed model, connect the saved training artifacts to the web interface, and test the end-to-end prediction workflow. We also need to refine the sensitivity analysis and feature summary panels, complete debugging, and polish the final report. Our plan is to finalize the offline modeling first, then complete deployment and interface testing, and finally focus on documentation and final submission.